

A Multilingual Speech Translation System

Sponsor: Student led, no sponsor

Background:

With an increase in our world becoming more connected and collaborative, English is becoming one the most used languages worldwide. In fact, it is considered one of the fastest growing languages in the world. There is a need to improve communication between people who speak different languages. The goal of this project is to design and develop a real-time language translation system that connects to wireless earbuds and translates spoken language into English.

The system will be implemented in Python and will use OpenAI models for speech-to-text and text-to-speech processing. It will use multiprocessing so that the system can continue listening for new speech while it is still translating previous speeches.

The hardware aspect will be created using a Raspberry pi so that we can implement code that can then turn the detected speech into a translated output. The ear piece must implement a speaker to output the inputted speech. This piece of technology also requires the use of a button that begins to record input data. This piece of technology will have a frequency adjusting system that allows comfort and a clear reading for the client.

Project Needs

- A system that can connect to and route audio through an ear piece.
- A speech recognition system that can detect and transcribe speech in different languages.
- Automatic language detection to determine the language being spoken, without requiring the user to specify it in advance.
- Hardware to make the ear piece: A button that starts to record the sounds of the speaker must be implemented so that data can be collected.
- Implement a hardware device that can output the result of the translation. A speaker must be used so that this output can be heard.

Project Scope

- Capture spoken audio from a connected microphone and process the incoming audio
- Use a speech-to-text model to convert the spoken language into text.
- Translate the transcribed speech from the spoken language into English.
- Use a text-to-speech model to output the translated speech to the user through the connected earbuds.

Potential Technical Considerations

- Use OpenAI's audio APIs (speech-to-text and text-to-speech) for transcription, language detection, and translation.

- Use Python multiprocessing so audio capture, translation, and speech output can run concurrently.
- Use an audio I/O library like PyAudio to capture microphone input and play back translated audio.
- Use a raspberry pi to download code to make the earpiece collect data
- Processing levels in order to keep up with the inputted data
- Signal processing: adjustment bands to adjust the frequency of audio grams
- Feedback cancellation: detect and block unwanted frequencies that will interfere with the inputted signals

Deliverables

- A working language translation system that translates spoken language into English and returns the translated speech to the user through connected wireless earbuds.

Dependencies:

TBD